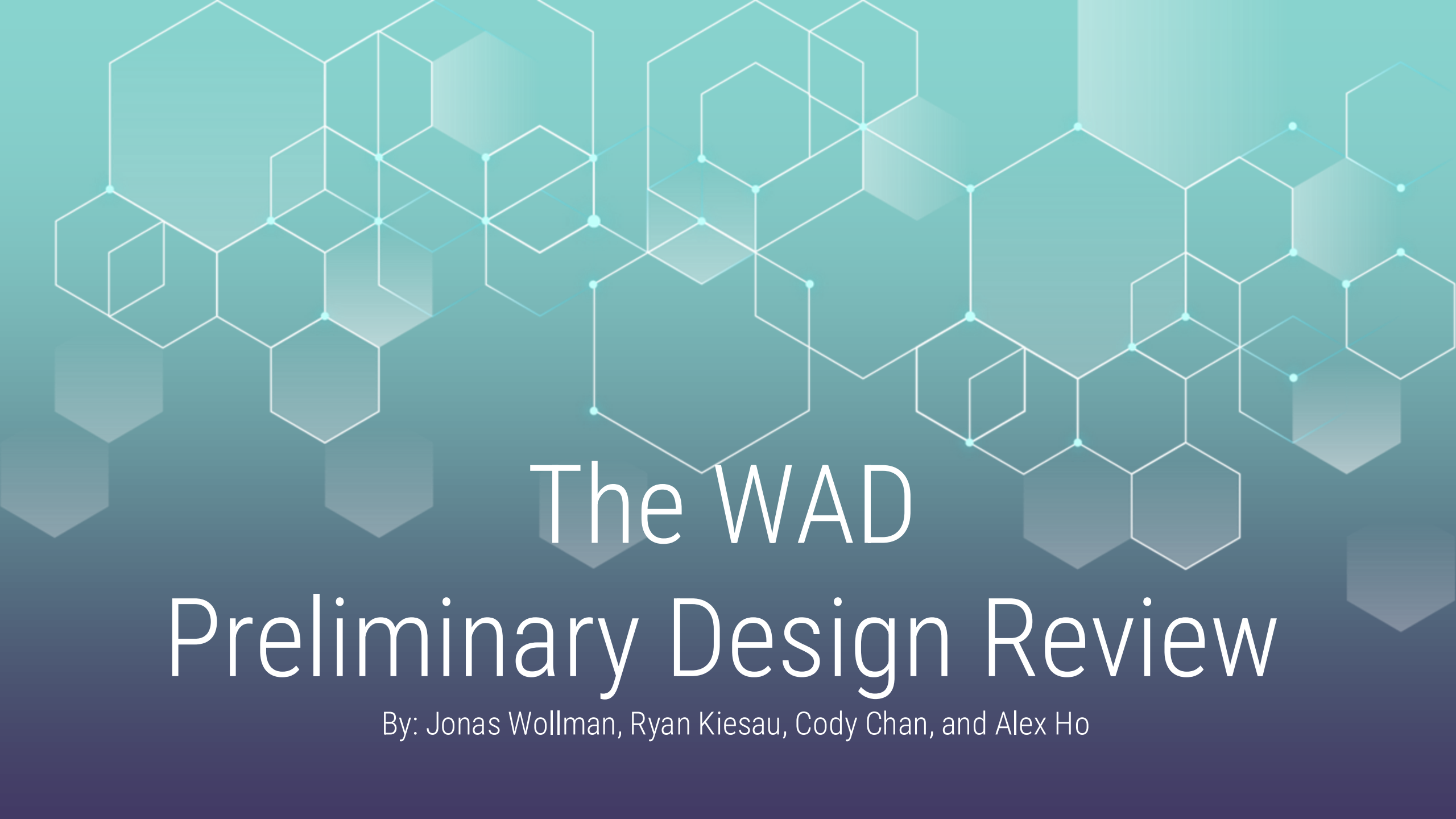




The WAD

Preliminary Design Review

By: Jonas Wollman, Ryan Kiesau, Cody Chan, and Alex Ho

Introductions



Name: Jonas Wollman

Major: Computer Engineering

Role: Project Manager

Post-College Plans: Working as an electronics engineer for the 76th Software Engineering Group (555th Squadron) at Tinker Air Force Base



Introductions



Name: Ryan Kiesau

Major: Electrical Engineering

Role: Hardware

Post-College Plans: Working as an electronics engineer for the 76th Software Engineering Group (557th Squadron) at Tinker Air Force Base

Introductions



Name: Cody Chan

Major: Computer Engineering

Role: Hardware

Post-College Plans: Pursue
Master's in Electrical Engineering
-The University of Oklahoma

Introductions



Name: Alex Ho

Major: Computer Engineering

Graduation Date: May 2026

Role: Software

Post-College Plans: Working as an electronics engineer at Tinker Air Force Base

Problem & Plan

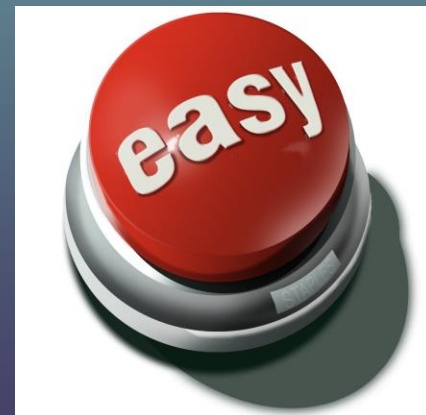
1. Wearability



2. Functionality

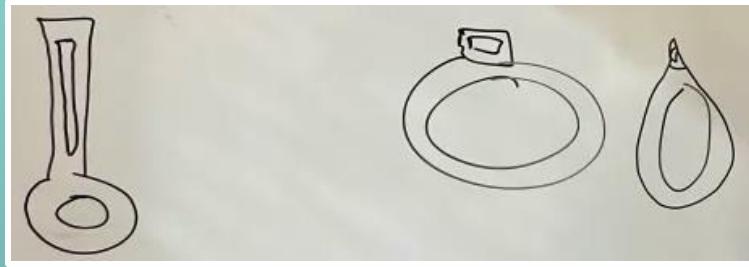


3. Ease of Use



Problem & Plan

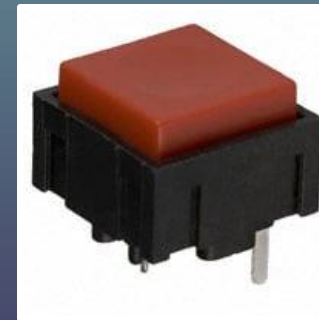
1. Pager



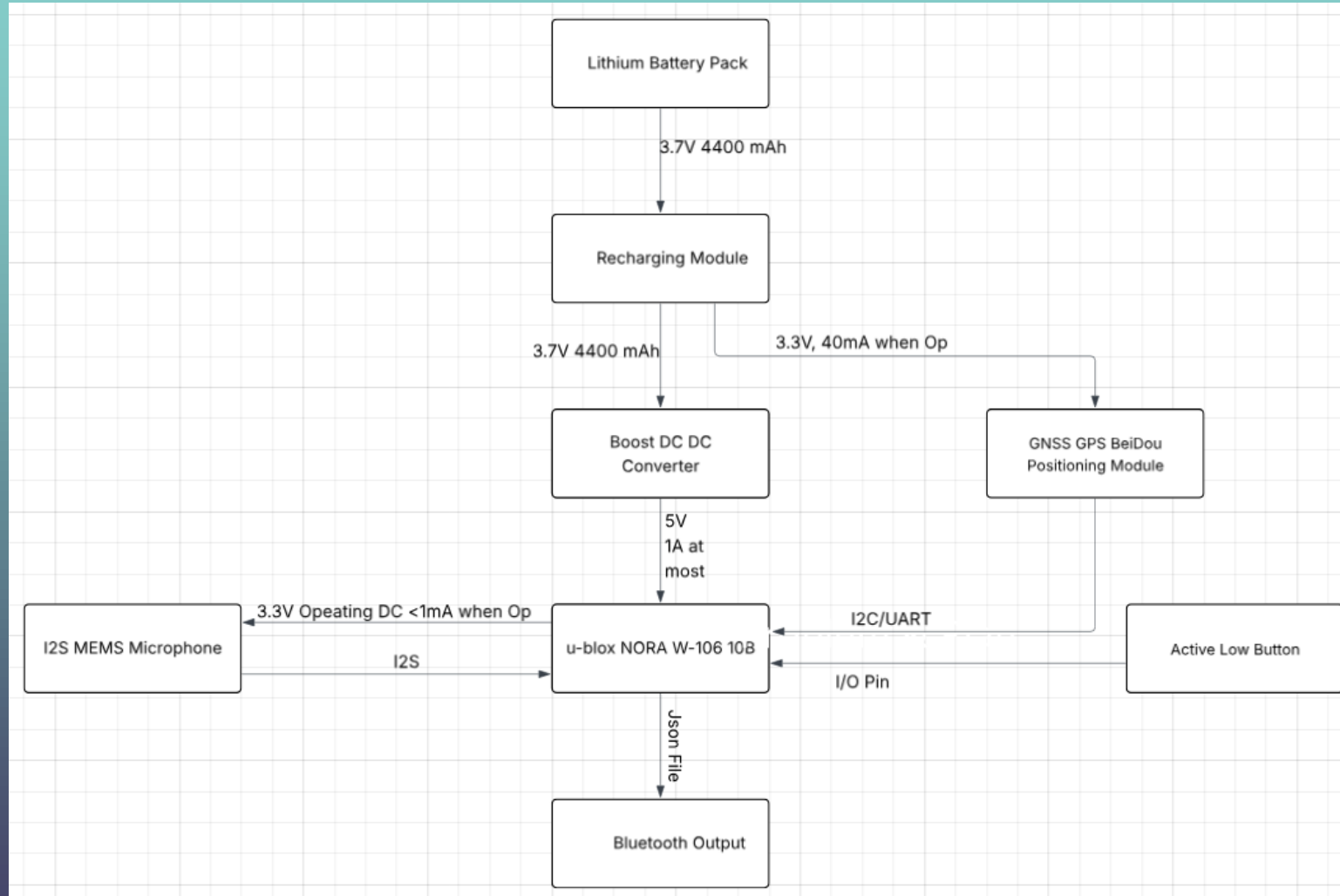
2. Gunshot & GPS



3. Click of a Button

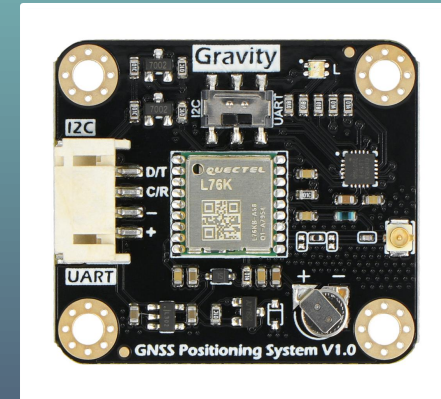


Overview Block Diagram

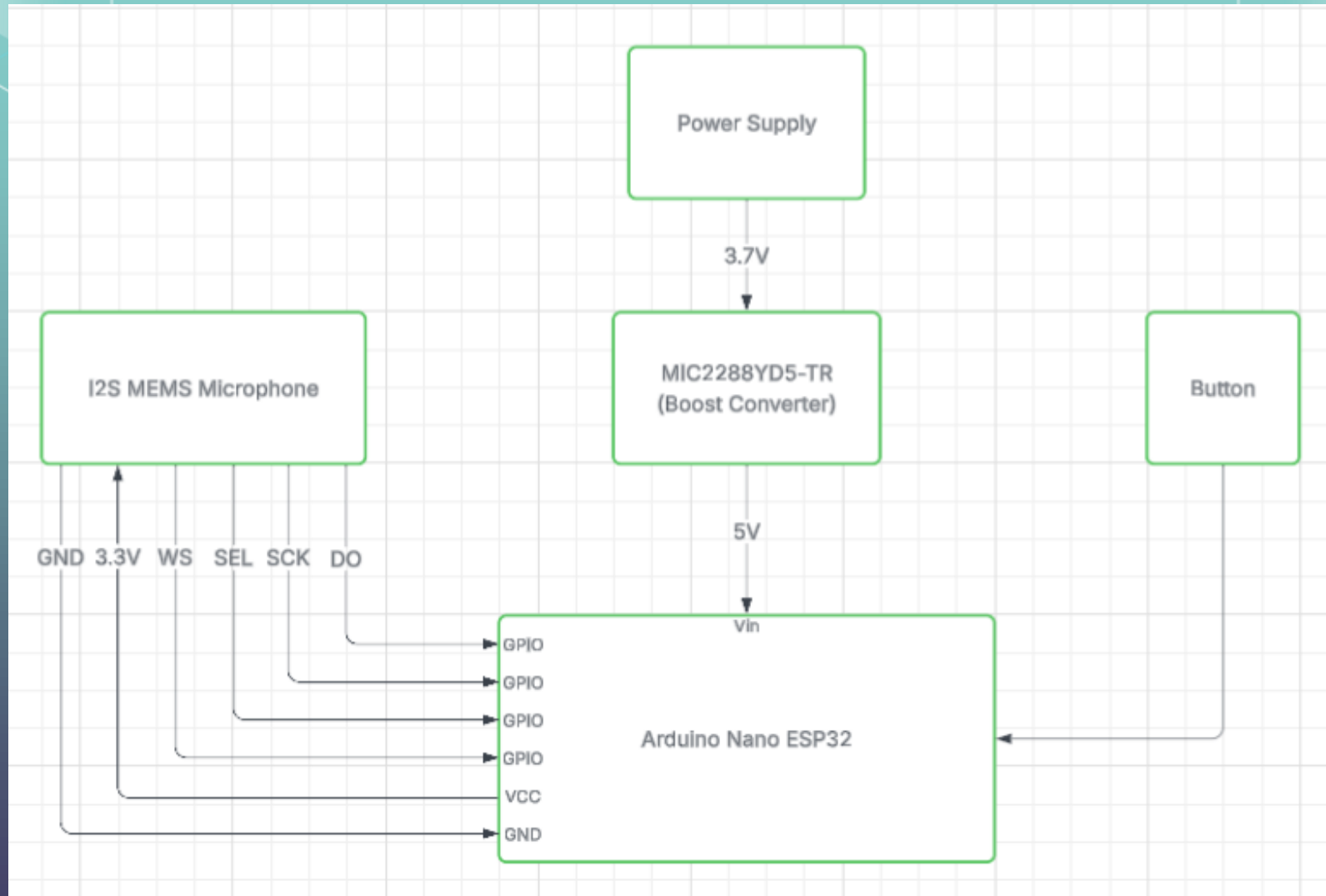


Hardware System Overview

- Main components:
 - Arduino Nano ESP32
 - Uses U-Blox NORA Chip W-106 10B
 - Bluetooth 5.0 standards (Bluetooth Classic and Bluetooth Low-Energy)
 - I²S MEMS Microphone
 - GPS Module
 - Push Button
 - Lithium Rechargeable Battery



Audio Sampling Block Diagram



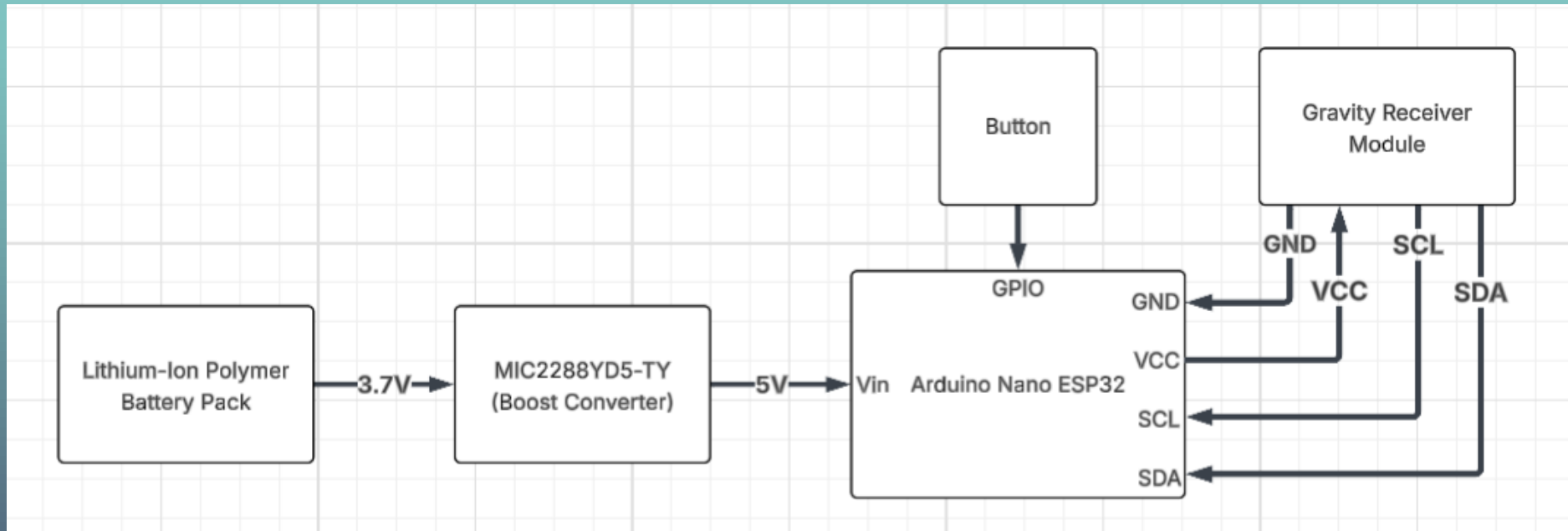
Microphone Hardware

- Fermion I2S MEMS Microphone
 - Operating Voltage: 3.3V
 - Low Current: < 1mA
 - Sound Pressure Level: 140 dB
 - Signal-to-Noise Ratio: 59 dB
 - Supports I²S



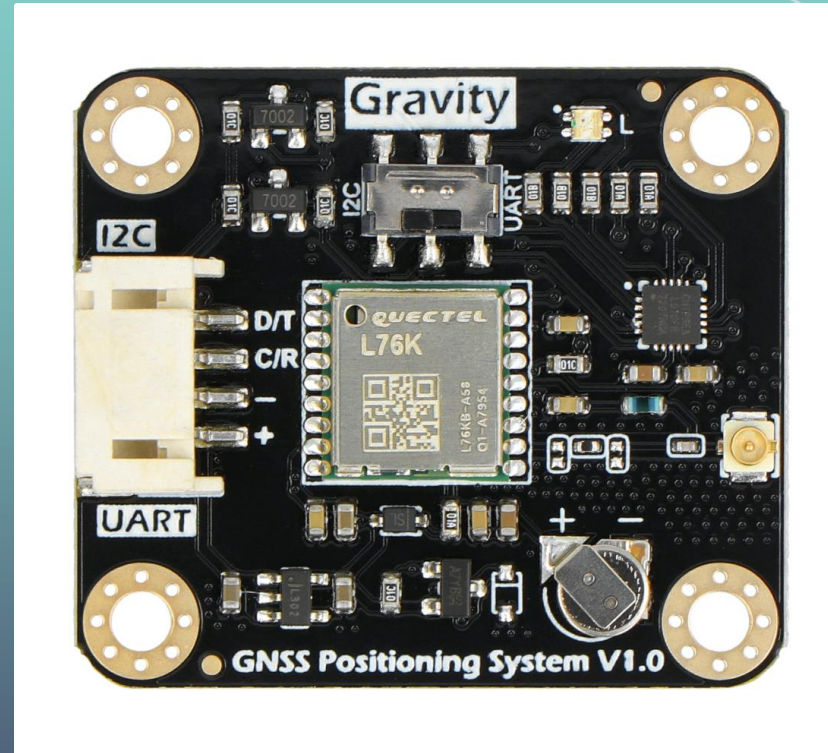
Fermion I2S MEMS Microphone

Positioning Block Diagram



Positioning Hardware

- Gravity Receiver Module
 - Multi-Satellite System
 - Supports I²C
 - 3.3V, 40mA
 - 30ns Positioning, 2m Accuracy
 - L1 and B1 Frequency Range
 - Offers Altitude Tracking
 - Plug and Play



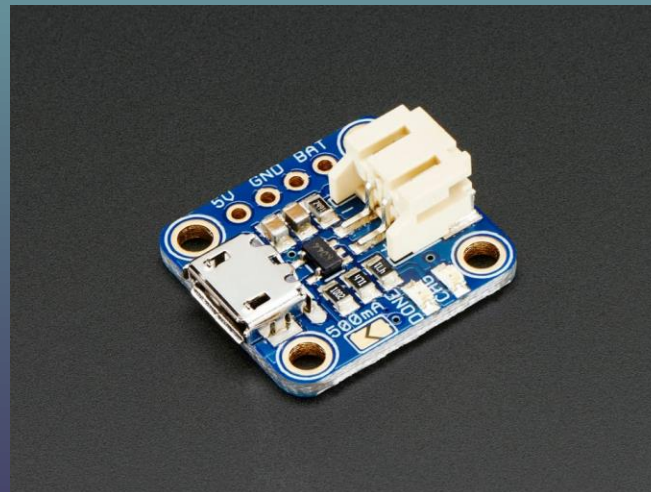
Gravity Receiver Module

Power Supply

- Lithium-Ion Polymer Battery Pack
 - 3.7V
 - 4400mAh
 - Can run entire device for 16 hours
- Micro-Lipo Charger
 - 5V Input
 - Micro-B USB Connector
 - 100mA Charge Current



Lithium-Ion Polymer Battery Pack



Micro-Lipo Charger

Software Design

Overview of Implementation

Programming Language: C++11 (follows joint technical standard ISO¹/IEC² 14882)

Main Functionality:

1. Collect audio samples and process to find gunshots
2. Collect GPS data
3. Package data and transmit to mobile app



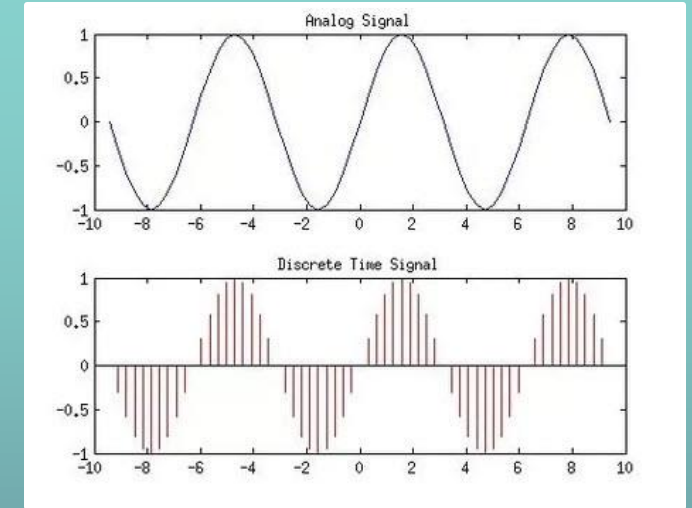
1. ISO – International Organization for Standardization

2. IEC – International Electrotechnical Commission

Software Design

Audio Signal Collection

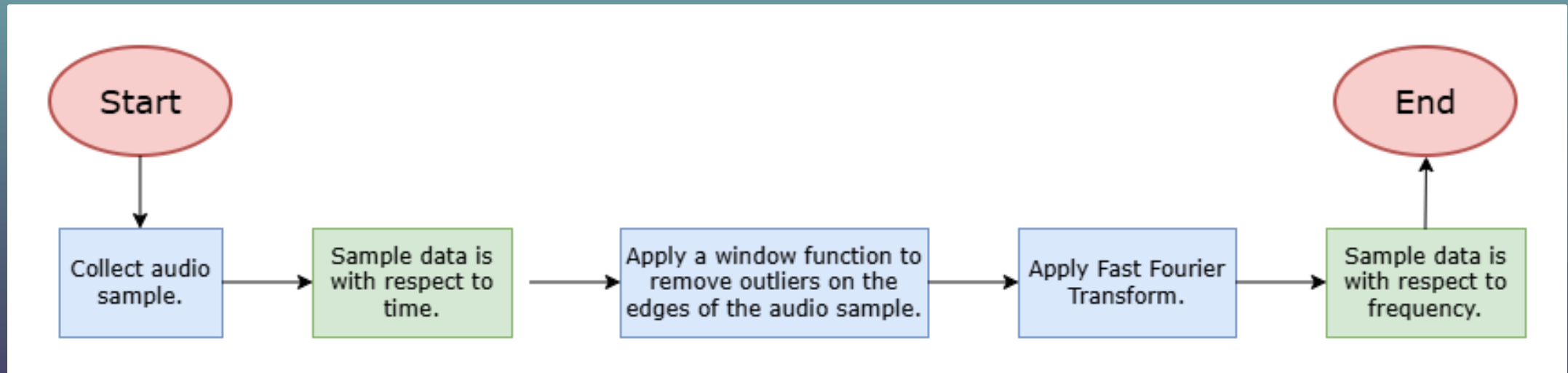
- Data received from microphone in digital PCM¹ format
 - I²S protocol to sample audio from microphone
 - Sampling Rate: 44.1 kHz
-
- Standard frequency for recreating audio signal with the same quality
 - ESP32 supports sampling PCM¹ audio at up to 100 kHz
 - ESP32 has enough memory to hold large set of audio data (512 KB of SRAM memory)



Signal Processing Design

Audio Signal Processing

- Analyzing audio sample with respect to frequency rather than time gives more information on gunshot audio patterns
- Software Library: arduinoFFT by @kosme on GitHub
- Aiming for at least 75% accuracy with gunshot detection



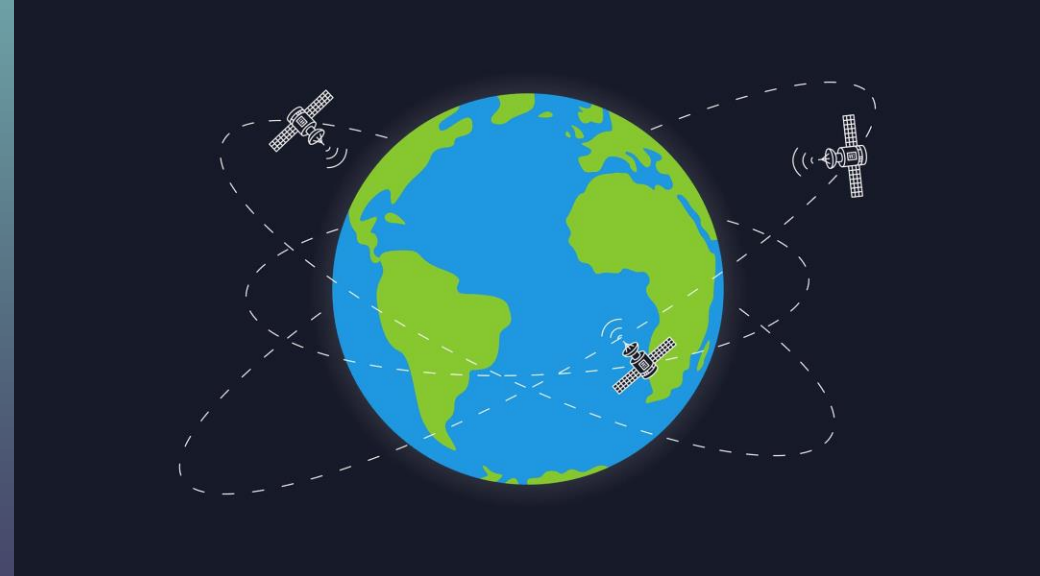
GPS Data Processing

GPS Data Processing

- Obtains GPS data using I²C (Inter-Integrated Circuit) protocol
- Sampling Rate: 1 Hz

Collected Data:

1. Timestamp (CST)
2. Latitude (degrees)
3. Longitude (degrees)



Bluetooth Design

Wireless Data Transmission

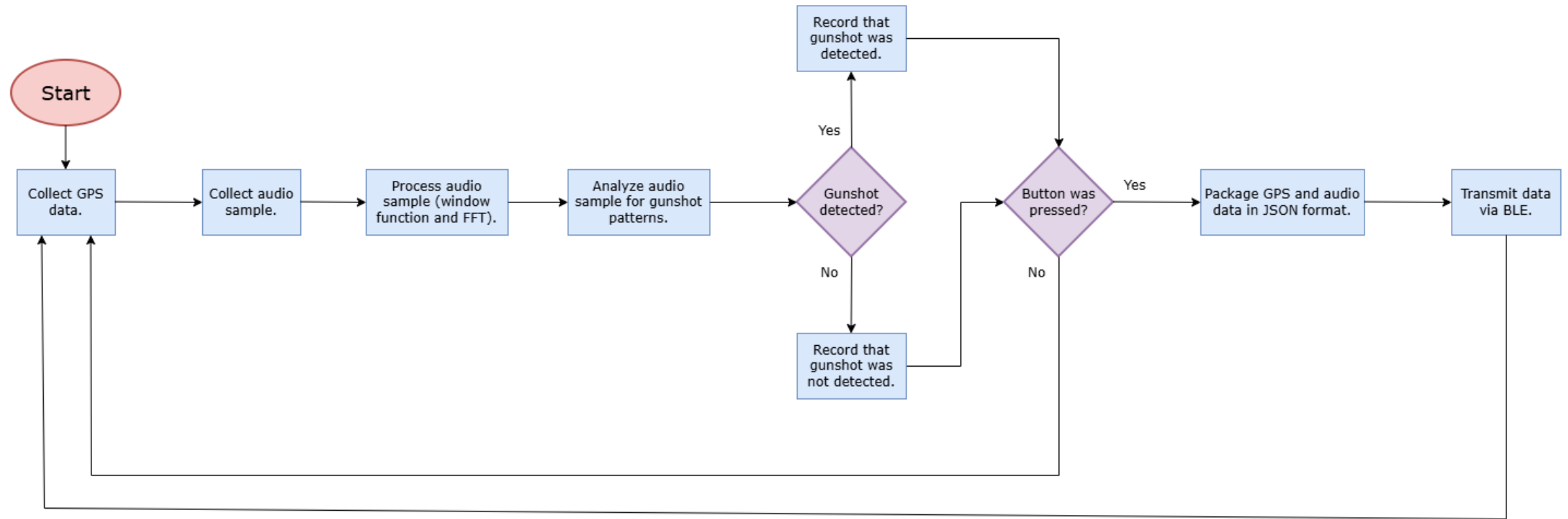
- Audio and GPS data will be packaged together in JSON file and transmitted via Bluetooth Low-Energy (BLE)

Testing:

- Since mobile app is not functional yet, data will be transmitted to a completely different microcontroller (another Arduino Nano ESP32) to see if transmission is successful



Software Block Diagram



Budget Plan

Item	Item Description	Quantity	Cost \$	Use	Part Number	Link
1	Arduino Nano ESP32	1	\$19.30	Main Micro controller for project	Mouser - ABX00083	https://www.mouser.com/ProductDetail/Arduino/ABX00083?qs=ulEaXIWI0c8OkA6u7pnNRA%3D%3D
2	Lithium Ion Battery Pack - 3.7V 4400mAh	1	\$19.95	Main Power supply for project	Adafruit - 354	https://www.adafruit.com/product/354
3	Micro-Lipo Charger for LiPoly Batt with USB Type C Jack	1	\$5.95	Rechargable module for Battery Pack	Adafruit - 4410	https://www.adafruit.com/product/4410
4	Gravity: GNSS GPS BeiDou Receiver Module - I2C&UART	1	\$17.90	GPS Module for Project	Dfrobot - 2651	https://www.dfrobot.com/product-2651.html
5	Fermion: I2S MEMS Microphone (Breakout)	1	\$4.90	Microphone Module for GS Detection	Dfrobot - 2637	https://www.dfrobot.com/product-2637.html
6	320E11RED	2	\$2.09	Button for panic device	Digikey - 101391	https://www.digikey.com/en/products/detail/e-switch/320E11RED/101391
7	MIC2288YD5-TR	1	\$0.73	Booster DC DC Converter	Mouser - MIC2288YD5	https://www.mouser.com/ProductDetail/Microchip-Technology/MIC2288YD5-TR?qs=Y3Q3JoKAO1S0eR8M4KE3QA%
		Total:	\$70.82			

Upcoming Plans

1-2 Weeks from Today:

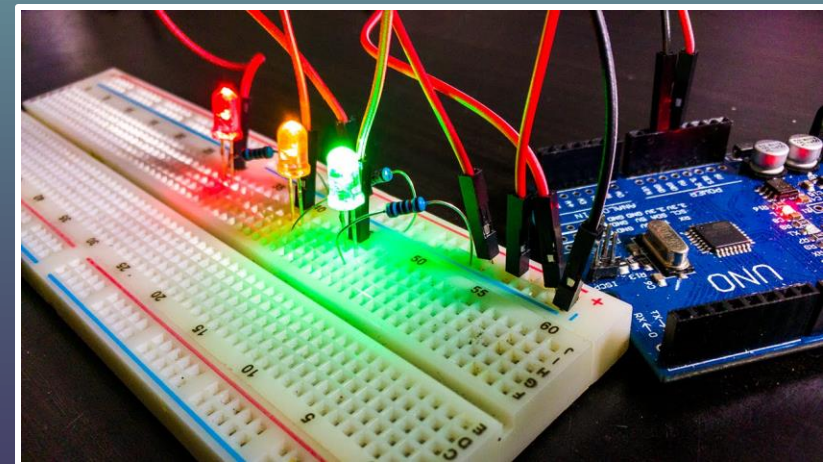
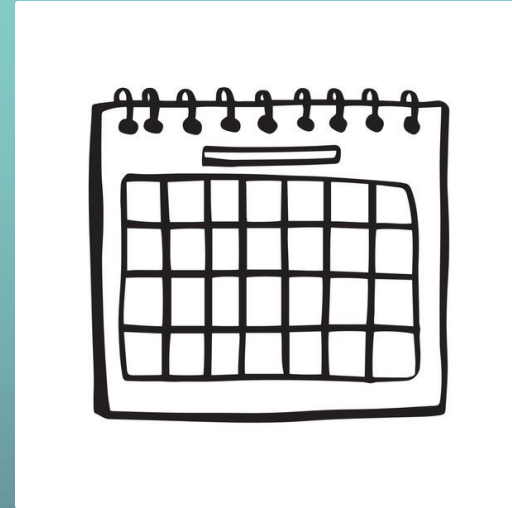
- Ordering Parts & Creation of Schematics

2-4 Weeks from Today:

- Building of Board & Debugging

4-8 Weeks from Today:

- Testing & Future Planning



Questions?